

CS 6341 Robotics Project Proposal Description

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1 Introduction

For the robotics course project, students can choose a topic related to robotics and explore the topic in one of the three different ways:

- **Research-oriented.** In this direction, students are going to propose a new idea that has not been explored before in the literature, then implement the new idea and conduct experiments to verify it.
- **Application-oriented.** In this direction, students can apply an existing robotics algorithm or method to a new problem or a new application. For example, if a method is proposed for domain A, the project can explore applying the method to domain B where different data are collected.
- **Implementation-oriented.** In this direction, students can select an existing robotics algorithm or method, and then implement it and perform experiments to verify the implementation. Since most robotics methods are open source these days, **for implementation-oriented projects, students cannot just use an open-source code and run experiments with it.**

For project evaluation, all three categories will be considered equally. A project will be evaluated according to its quality in terms of implementation, experiments, presentation, and writing, regardless of its category. However, **students are encouraged to consider research-oriented projects and application-oriented projects.** Even if the novelty introduced is incremental, it is still exploring new things researchers have not tried before or applying an approach to new applications.

2 Proposal Submission (5%)

The project proposal should be prepared using the ICRA double column latex format. A useful online LaTeX tool is Overleaf <https://www.overleaf.com/>. We have the ICRA latex template accessible here through the overleaf: <https://www.overleaf.com/read/rwmhwnwjkrmc>. You can download a copy of the template or make a copy on the overleaf for your own project, and then edit it. **Please make sure that you use this latex template for your project. Otherwise, some point will be deducted.**

The project proposal should be **1-page PDF** using the latex template with the following items:

- **Title.** Let us give a name to your project.
- **Team Members.** List the names of the team members as authors in the proposal.
- **Problem Statement.** Describe the problem you are trying to solve in this project.
- **Approach.** Describe your idea to solve the problem. It is fine if some details have not been determined in the project proposal. But students should have rough ideas of how to proceed.

First, explicitly state which category the project is in: research-oriented, application-oriented, or implementation-oriented. Second, for research-oriented projects, describe the proposed idea and the novelty of the idea. For application-oriented projects, describe which approach is going to be used and how to apply this approach to a new application. For implementation-oriented projects, describe which approach is to be implemented and the plan for implementation.

- **Simulation Environments (Optional).** Describe the simulation environment that will be used in the project for robot experiments.
- **Evaluation.** Describe how to evaluate the success of the project. For example, what evaluation metrics will be used to evaluate the performance of the method?
- **References.** Cite related works in the proposal.

There is one mandatory requirement for the course project:

- **The project needs to use the SO-101 arm, and demonstrate a task that the arm can do.**

3 Proposal Video Submission (5%)

After calibrating the SO-101 arm, each team must contact the TA to test teleoperation using the team's leader and follower arms. The TA will help verify that the arms are properly calibrated and configured for teleoperation.

Each team must submit a video demonstrating successful teleoperation. The video must clearly show:

- The leader and follower SO-101 arms used by the team.
- The follower arm correctly following the motions of the leader arm through teleoperation.
- Multiple joint motions that demonstrate that the robot is properly calibrated and configured.

The video should be recorded after the teleoperation setup has been tested with the TA. Successful teleoperation will demonstrate that the SO-101 arms have been correctly calibrated and are ready for use in the course project.

4 Suggested Topics

Based on the materials we cover in the lectures, we suggest the following topics for the course project. However, the scope of the project is not limited to the topics mentioned below. **Students can explore other topics in robotics as well.** Also, the references in the suggested topics are recent representative works. **Students can explore methods beyond these references and propose new ideas for different topics.**

- **Model-based Grasping.** Using 3D models of objects, we can first estimate the 6D object pose, i.e., 3D rotation and the 3D translation of objects [35, 27, 29, 5]. Then model-based grasp planning [20, 6] and motion planning (e.g., MoveIt [2]) approaches can be used for robot grasping.
- **Model-free Grasping Planning.** If we cannot get 3D models of objects but we can obtain images or 3D point clouds of objects using object segmentation methods [36, 18, 14], grasps can be planned using images or point clouds [19, 4, 21, 26].
- **Reinforcement Learning for Grasping.** Reinforcement learning approaches have been widely studied for robot grasping. Most approaches focus on the use of RL for top-down grasping [22, 12, 10, 28, 41]. A few works apply RL to 6D grasping [24, 31, 30].
- **Manipulation of Articulated or Deformable Objects.** In addition to manipulation of rigid objects, active research interests focus on manipulation of articulated objects [13, 34, 17, 37] and deformable objects [40, 23, 15, 16].
- **Mobile Manipulation.** In this case, a robot needs to navigate and manipulate objects [11, 25, 7, 9, 33]. Mobile manipulation is suitable for robots working in large environments.
- **Language-guided Manipulation.** Humans can use natural language to instruct robots to perform manipulation tasks [1, 8, 32].
- **Human-Robot Handover.** Robots and humans can work together to perform tasks. Human-robot handover is a simple example in which humans hand objects to robots or vice versa [38, 31, 39, 3].

5 Robot Simulator Resources

Here are a few robot simulators and simulation environments that you can use for the course project.

- Gazebo <https://gazebo.org/home>. Gazebo is integrated with ROS. If your code runs in Gazebo, it can be easily transferred to a real robot.
- PyBullet <https://pybullet.org/wordpress/>. PyBullet is an easy-to-use simulator with Python interfaces.
- NVIDIA Isaac Lab <https://developer.nvidia.com/isaac/lab>. NVIDIA Isaac™ Lab is an open-source, unified framework for robot learning designed to help train robot policies.

- iGibson <https://svl.stanford.edu/igibson/>. iGibson is a simulation environment based on PyBullet. It can be used for robot manipulation and navigation.
- AI2-THOR <https://ai2thor.allenai.org/>, a simulation environment for navigation.
- Habitat <https://aihabitat.org/>, a simulation environment for embodied AI.
- SAPIEN <https://sapien.ucsd.edu/>. SAPIEN contains asserts of articulated objects. It can be useful for manipulating articulated objects.
- BulletArm <https://github.com/ColinKohler/BulletArm>, a PyBullet-based simulation environment for benchmarking of several robot manipulation tasks.

6 Deep Learning Resources

The most recent vision methods leverage deep learning to train neural networks to tackle various problems in robotics. If your project requires deep neural network training, you may need GPUs for training. **Google Colab** is a great free resource for small amounts of GPU resources: <https://colab.research.google.com/>. Two widely-used deep learning frameworks:

- PyTorch <https://pytorch.org/>
- TensorFlow <https://www.tensorflow.org/>

7 Resources from IRVL at UT Dallas

Our lab, Intelligent Robotics and Vision Lab (IRVL) at UT Dallas, provides several software that can be useful for your projects:

- The SceneReplica benchmark for robot grasping <https://irvlutd.github.io/SceneReplica/>. This project contains links to several object perception, grasp planning, and motion planning software.
- The VLA-Replica benchmark for VLA models <https://irvlutd.github.io/VLAReplica/>.
- Grasping trajectory optimization <https://github.com/IRVLUTD/GraspTrajOpt>
- Isaac Sim for grasping <https://irvlutd.github.io/MultiGripperGrasp/>
- GraspIt! <https://github.com/IRVLUTD/neuralgrasps-dataset-generation/tree/multigrippergrasp>
- Reinforcement learning for 6D grasping <https://sites.google.com/view/gaddpg>

You can find more in <https://labs.utdallas.edu/irvl/resources/>.

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